4.7.1 Forces and motion

Forces can change the motion of objects. Objects can move in a straight line at a constant speed. They can also change their speed and/or direction (accelerate or decelerate). This topic shows how observations of moving objects can be accounted for in terms of Newton's laws of motion. Graphs can help to describe the movement of objects. These may be distance–time or velocity–time graphs. When an object speeds up or slows down, its kinetic energy increases or decreases. The forces that cause the change in speed do so by doing work. The topic ends by showing that these ideas can be applied to the issue of road safety.

The required practical in this topic is an investigation of the effects of varying the force and/or mass on the acceleration of an object.

In this topic students have to be able to apply formulae relating distance, time and speed, for uniform motion and for motion with uniform acceleration, and calculate average speed for non-uniform motion (MS 1a, 1c, 2b, 3c).

4.7.1.1 Speed and velocity

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the vector–scalar distinction as it applies to displacement, distance, velocity and speed.	Distance is how far an object moves. Distance does not involve direction. Distance is a scalar quantity. Displacement includes both the distance an object moves, measured in a straight line from the start point to the finish point, and the direction of that straight line. Displacement is a vector quantity. Speed does not involve direction. Speed is a scalar quantity. The velocity of an object is its speed in a given direction. Velocity is a vector quantity.	

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall typical speeds encountered in everyday experience for wind and sound, and for walking, running, cycling and other transportation systems.	The speed of a moving object is rarely constant. When people walk, run or travel in a car their speed is constantly changing. The speed that a person can walk, run or cycle depends on many factors, including age, terrain, fitness and distance travelled. Typical mean values are: • walking 1.5 m/s • running 3 m/s • cycling 6 m/s. It is not only moving objects that have varying speed. The speed of sound and the speed of the wind also vary. A typical value for the speed of sound is 330 m/s.	

4.7.1.2 Distance, speed and time

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Make measurements of distances and times, calculate speeds, and make and use graphs of these to determine the speeds and accelerations involved.	The distance travelled by an object moving at constant speed increases with time. $\textit{distance travelled} = \textit{speed} \times \textit{time}$ $s = v\,t$ distance, s, in metres, m speed, v, in metres per second, m/s time, t, in seconds, s If an object moves along a straight line, how far it is from a certain point can be represented by a distance–time graph. The speed of an object can be calculated from the gradient of its distance–time graph. (HT only) If an object is accelerating, its speed at any particular time can be determined by drawing a tangent and measuring the gradient of the distance–time graph at that time.	MS 3b, 3c Recall and apply this equation. WS 4.5, MS 1c, 3b, 3c Use ratios and proportional reasoning to convert units and to compute rates. WS 1.2, 3.5, MS 4a, 4b, 4c, 4d, 4f Relate changes and differences in motion to appropriate distance–time, and velocity–time graphs, and interpret lines, slopes and enclosed areas in such graphs. MS 1a, 1c, 2f, 3c Calculate average speed for non-uniform motion.

4.7.1.3 Circular motion (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain with examples that motion in a circular orbit involves constant speed but changing velocity (qualitative only).	The velocity of an object is its speed in a given direction. Velocity is a vector quantity. When an object moves in a circle the direction of the object is continually changing. This means that an object moving in a circle at constant speed has a continually changing velocity.	

4.7.1.4 Free fall

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the acceleration in free fall and estimate the magnitudes of everyday accelerations.	Acceleration is the rate at which the velocity of an object changes. $acceleration = \frac{change\ in\ velocity}{time\ taken}$ $a = \frac{\Delta v}{t}$ acceleration, *a*, in metres per second squared, m/s^2 change in velocity, Δv, in metres per second, m/s time, *t*, in seconds, s An object that slows down (decelerates) has a negative acceleration. The acceleration of an object can be calculated from the gradient of a velocity–time graph. The distance travelled by an object (or displacement of an object) can be calculated from the area under a velocity–time graph.	WS 1.2, 3.3, MS 3b, 3c Recall and apply this equation. WS 1.2, 3.5, MS 4a, 4b, 4c, 4d, 4f, 5c Relate changes and differences in motion to appropriate velocity–time graphs, and interpret lines and slopes to determine acceleration. (HT only) Interpret enclosed areas in such graphs to determine distance travelled (or displacement). (HT only) Measure, when appropriate, the area under a velocity–time graph by counting squares.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
	(HT only) The following equation applies to uniform motion: $(final\ velocity)^2 - (initial\ velocity)^2 = 2 \times acceleration \times distance$ $\left[v^2 - u^2 = 2\ a\ s\right]$ final velocity, *v*, in metres per second, m/s initial velocity, *u*, in metres per second, m/s acceleration, *a*, in metres per second squared, m/s^2 distance, *s*, in metres, m Near the Earth's surface any object falling freely under gravity has an acceleration of about 9.8 m/s^2. An object falling through a fluid initially accelerates due to the force of gravity. Eventually the resultant force will be zero and the object will move at its terminal velocity.	WS 1.2, 3.3, MS 3c (HT only) Apply this equation, which is given on the *Physics equation sheet*.

4.7.1.5 Newton's First Law

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Apply Newton's First Law to explain the motion of objects moving with uniform velocity and also objects where the speed and/or direction change.	Newton's First Law: If the resultant force acting on an object is zero and: • the object is stationary, the object remains stationary • the object is moving, the object continues to move at the same speed and in the same direction. So the object continues to move at the same velocity.	

4.7.1.6 Newton's Second Law

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Apply Newton's Second Law in calculations relating forces, masses and accelerations.	Newton's Second Law: The acceleration of an object is proportional to the resultant force acting on the object, and inversely proportional to the mass of the object. $resultant\ force = mass \times acceleration$ $F = m\,a$ force, F, in newtons, N mass, m, in kilograms, kg acceleration, a, in metres per second squared, m/s^2	MS 3a Recognise and be able to use the symbol: • for proportionality, α • that indicates an approximate value or answer, ~ WS 1.2, 3.3, MS 3c Recall and apply this equation.
(HT only) Explain that inertial mass is a measure of how difficult it is to change the velocity of an object and that it is defined as the ratio of force over acceleration.	(HT only) The tendency of objects to continue in their state of rest or of uniform motion is called inertia. Inertial mass is a measure of how difficult it is to change the velocity of an object. Inertial mass is defined by the ratio of force over acceleration.	

Required practical activity 14: investigate the effect of varying the force on the acceleration of an object of constant mass and the effect of varying the mass of an object on the acceleration produced by a constant force.

AT skills covered by this practical activity: physics AT 1, 2 and 3.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 170).

4.7.1.7 Newton's Third Law

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall Newton's Third Law and apply it to examples of equilibrium situations.	Newton's Third Law: Whenever two objects interact, the forces they exert on each other are equal and opposite.	

4.7.1.8 Momentum (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Define momentum and describe examples of momentum in collisions.	Momentum is a property of moving objects. Momentum is defined by the equation: $momentum = mass \times velocity$ $p = m\,v$ momentum, p, in kilograms metre per second, kg m/s mass, m, in kilograms, kg velocity, v, in metres per second, m/s In a closed system, the total momentum before an event is equal to the total momentum after the event. This is called conservation of momentum.	WS 1.2, 3.3, MS 3c Recall and apply this equation. WS 1.2 Use the concept of momentum as a model to analyse an event such as a collision.

4.7.1.9 Kinetic energy

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate the amounts of energy associated with a moving body. Describe all the changes involved in the way energy is stored when a system changes for common situations: a moving object hitting an obstacle, or an object being accelerated by a constant force.	The kinetic energy of a moving object depends on the mass and the velocity of the object. $kinetic\ energy = 0.5 \times mass \times (speed)^2$ $\left[E_k = \frac{1}{2}\,m\,v^2\right]$ kinetic energy, E_k, in joules, J mass, m, in kilograms, kg speed, v, in metres per second, m/s	WS 1.2, 3.3, MS 3c Recall and apply this equation.

4.7.1.10 Stopping distances

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the factors which affect the distance required for road transport vehicles to come to rest in emergencies and the implications for safety.	The stopping distance of a vehicle is the sum of the distance the vehicle travels during the driver's reaction time (thinking distance) and the distance it travels under the braking force (braking distance). For a given braking force the greater the speed of the vehicle, the greater the stopping distance. The braking distance of a vehicle can be affected by wet or icy weather and poor condition of the vehicle's brakes or tyres. A driver's reaction time (see The human nervous system (page 41)) can be affected by tiredness, drugs and alcohol. Distractions may also affect a driver's ability to react.	WS 3.6 Analyse a given situation to explain why braking could be affected. WS 1.5 Discuss the implications for safety. WS 3.5, MS 4a Interpret graphs relating speed to stopping distance for different types of vehicles. WS 1.5, 2.2, MS 1a, 1c Evaluate the effect of various factors on thinking distance based on given data.
Explain the dangers caused by large decelerations. Describe all the changes involved in the way energy is stored when a system changes, for common situations, like a vehicle slowing down.	When a force is applied to the brakes of a vehicle, work done by the friction force between the brakes and the wheel reduces the kinetic energy of the vehicle and the temperature of the brakes increases. The greater the speed of a vehicle the greater the braking force needed to stop the vehicle in a certain distance. The greater the braking force the greater the deceleration of the vehicle. Large decelerations may lead to brakes overheating and/or loss of control.	WS 1.5, MS 1d (HT only) Estimate the forces involved in typical situations on a public road.